Introduction to JDBC (Java Database Connectivity)

Section 1: Learn

What is JDBC?

JDBC stands for **Java Database Connectivity**. It is a **Java API** that allows Java programs to **interact with relational databases** like MySQL, Oracle, PostgreSQL, etc.

JDBC helps Java applications to **send SQL statements** to a database and **receive results**.

Purpose of JDBC

- **Connect Java applications** to a database
- Perform **CRUD operations** (Create, Read, Update, Delete)
- Enables **platform-independent** database access
- Acts as a **bridge between Java code and SQL commands**

Common JDBC Operations

- Establishing a connection to the database
- Creating SQL statements
- Executing SQL queries
- Processing the results
- Closing the connection

Types of JDBC Drivers

Java uses different **drivers** to connect with databases. These drivers are categorized into **4 types**:

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1. JDBC-ODBC Bridge Driver (Type 1)

- Connects Java to database via ODBC driver.
- Uses a native ODBC driver installed on the machine.

Advantages:

- Easy to use for learning

Disadvantages:

- Platform-dependent
- Deprecated in Java 8

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2. Native-API Driver (Type 2)

- Converts JDBC calls to native database API using native libraries.

Advantages:

- Faster than Type 1

Disadvantages:

- Requires native DB libraries
- Platform-dependent

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3. Network Protocol Driver (Type 3)

- Uses **middleware server** to convert JDBC calls into DB-specific protocol.

Advantages:

- Platform independent

- Good for intranet applications

Disadvantages:

- Requires middleware

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4. Thin Driver (Type 4)

- Pure Java driver that directly connects to the database over the network.

Advantages:

- Platform-independent
- Fast and efficient
- Most commonly used

Disadvantages:

- Database-specific (each DB has its own Type 4 driver)

Summary Table

Driver Type	Name	Platform Dependent	Usage Today
Type 1	JDBC-ODBC Bridge	Yes	Deprecated
Type 2	Native-API	Yes	Rare
Type 3	Network Protocol	No	Limited
Type 4	Thin Driver	No	Common

Section 2: Practice

Try Listing Driver Types

- Identify which driver is used by:
    - **MySQL** – Type 4
    - **Oracle** – Type 4
    - **Microsoft SQL Server** – Type 4

Section 3: Know More (FAQs)

Which JDBC driver is best?

Type 4 (Thin Driver) – It's platform-independent and fast.

Do I need to install drivers separately?

Some drivers (like MySQL Connector) need to be added as a **JAR file** to the project.

Is JDBC a library?

Yes. It's part of `java.sql` package in Java SE.

Can JDBC be used with any database?

Yes, as long as there's a JDBC driver available for that database.

Is JDBC still used today?

Yes. JDBC is a **core technology** for database access in Java, especially for **enterprise applications**.